Heisenberg Group and Reality

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1

Let H be the complex Heisenberg group, an extension of $\mathbb{C}^2$ by $\mathbb{C}^*$. It has coordinates (x, y, λ), the group law being

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \tag{1.1}$$

The involution

$$\tau : (x, y, \lambda) \longmapsto (-x, -y, \lambda) \tag{1.2}$$

is an automorphism of H.

The coordinate system $x, y, \lambda : H \to \mathbb{C}^2 \times \mathbb{C}^*$ exhibits the section $\mathbb{C}^2 \times \{1\}$ of the projection of H onto $\mathbb{C}^2$. On this section, $\tau(g) = g^{-1}$. Each line $D \subset \mathbb{C}^2$ has, by means of this section, a lift to a one-parameter subgroup of H, again denoted by D. The commutator $(u, v) = uvu^{-1}v^{-1}$ of two elements $u = (u_1, u_2)$ and $v = (v_1, v_2)$ of $\mathbb{C}^2$ is $\exp(2\pi i\langle u, v\rangle) \in \mathbb{C}^*$, with

$$\langle u, v\rangle = u_1v_2 - u_2v_1. \tag{1.3}$$

Let $H_{\mathbb{R}}$ be the real form defined by $x, y \in \mathbb{R}$ and $|\lambda| = 1$. The group $\mathrm{SL}(2, \mathbb{C})$ acts on H, and $\mathrm{SL}(2, \mathbb{R})$ acts on $H_{\mathbb{R}}$, which as a set is identified with $\mathbb{R}^2 \times U^1$.

It is known, by the Stone-von Neumann theorem [?], that $H_{\mathbb{R}}$ has a unique irreducible unitary representation V for which $\lambda \in U^1$ acts by multiplication by λ. In the Schrödinger model of V, one has $V = L^2(\mathbb{R})$, and the action of $H_{\mathbb{R}}$ is

$$\begin{aligned} (x, 0, 1) :&\quad f(z) \longmapsto f(z + x), \\ (0, y, 1) :&\quad f(z) \longmapsto e^{2\pi iyz} f(z). \end{aligned} \tag{1.4}$$

From the uniqueness of V one deduces a projective unitary action of $\mathrm{SL}(2, \mathbb{R})$ on V, compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $H_{\mathbb{R}}$ (A. Weil [?], D. Shale [?]).

For the action of $H_{\mathbb{R}}$ on $V = L^2(\mathbb{R})$, the C^∞ vectors, i.e. those $v \in V$ for which $h \mapsto hv$ is C^∞ in $h \in H_{\mathbb{R}}$, form the Schwartz space $\mathscr{S}$ of C^∞ rapidly decreasing functions on $\mathbb{R}$. The holomorphic vectors, i.e. those $v \in V$ for which $h \mapsto hv$ extends to a holomorphic map from H to V, are the L^2 functions on $\mathbb{R}$ that extend to holomorphic functions on $\mathbb{C}$ such that, in every horizontal strip $|\operatorname{Im} z| < A_1$, one has, for all $A_2 \in \mathbb{R}$,

$$|f(z)| \leq O\big(\exp(-A_2|z|)\big).$$

The complex group H acts on the space V^{hol} of holomorphic vectors by the formulae (1.4).

For the holomorphic action of H on V^{hol}, one does not have the same uniqueness as for the unitary action of $H_{\mathbb{R}}$ on V. If one did, the projective action of $\mathrm{SL}(2, \mathbb{R})$ on V would extend to a projective action of $\mathrm{SL}(2, \mathbb{C})$ on V^{hol}; since $\mathrm{SL}(2, \mathbb{C})$ is simply connected, this action would lift to an honest action of $\mathrm{SL}(2, \mathbb{C})$, and the projective action of $\mathrm{SL}(2, \mathbb{R})$ would lift to an honest action. This is not the case: only the double cover of $\mathrm{SL}(2, \mathbb{R})$ acts. Our purpose is to make explicit what happens.

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Let $S = \mathbb{P}^1(\mathbb{C})$ be the space of lines of $\mathbb{C}^2$. For $D \in S$, lifted in H as in no. 1, let W_D be the space of holomorphic functions φ on H satisfying

$$\varphi(\lambda dh) = \lambda\varphi(h)$$

for $\lambda \in \mathbb{C}^*$ and $d \in D$, endowed with the action of H given by $g : \varphi(h) \mapsto \varphi(hg)$. For $E \neq D$, restriction to E is a bijection

$$r_E : W_D \xrightarrow{\sim} \mathcal{O}(E), \tag{2.1}$$

where $\mathcal{O}(E)$ denotes the space of holomorphic functions on E.

If R is a holomorphic representation of H, on which each $\lambda \in \mathbb{C}^*$ acts by λ, and if ω is a D-invariant linear form on R, then

$$w \longmapsto \omega(hw) \tag{2.2}$$

is a morphism from R to W_D.

3. Example

Let V, as in nos. 1 and 2, be the irreducible unitary representation of $H_{\mathbb{R}}$, let $\mathscr{S} \subset V$ be the space of its C^∞ vectors, and let $\mathscr{S}' \supset V$ be the dual of the space of C^∞ vectors of the dual representation; in the Schrödinger model, this is the space of tempered distributions.

Let $D_t \in S$ be the line generated by $(-t, 1)$. If $\operatorname{Im} t > 0$, the linear form, written in the Schrödinger model,

$$\omega_t : f \longmapsto \frac{1}{\sqrt{t/i}} \int_{\mathbb{R}} f(x) \exp(-\pi i x^2/t)\, dx$$

is defined on V, and even on $\mathscr{S}'$. The form ω_t is a holomorphic vector of the dual of V, fixed by D_t. In terms of the infinitesimal generator $-t\partial_x + 2\pi i x$ of D_t, the invariance is written

$$\int_{\mathbb{R}} [(-t\partial_x + 2\pi i x) f(x)] \exp(-\pi i x^2/t)\, dx = \int_{\mathbb{R}} f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t)\, dx = 0.$$

By (2.2), ω_t defines

$$\Omega_t : V^{\mathrm{hol}} \longrightarrow W_{D_t},$$

which extends to V and even to $\mathscr{S}'$. Let E be the x-axis. The map $r_E\Omega_t : V \to \mathcal{O}(E)$ sends $f \in V$ to

$$f(x,t) = r_E\Omega_t(f) \qquad (\operatorname{Im} t > 0),$$

a solution of the heat equation

$$\left(\partial_t - \frac{1}{4\pi i}\partial_x^2\right) f(x,t) = 0.$$

For real t, the form ω_t remains defined on $\mathscr{S} \subset V$. For $t = 0$ it is $f \mapsto f(0)$, and for $t = \infty$ it is $f \mapsto \int f(x)\, dx$. The homomorphism Ω_t remains defined on V^{hol}. For $t = 0$, $r_E\Omega_0$ is, in the Schrödinger model, the identity.

Assume $\operatorname{Im} t > 0$. Let $\overline{D}_t$ be the complex conjugate of D_t. In the affine line $S - \{\overline{D}_t\}$, the "half-plane" $\{D_u \mid \operatorname{Im} u > 0\}$ is a disk with center D_t. The map $r_{\overline{D}_t} \circ \Omega_t : V \to W_{D_t} \to \mathcal{O}(\overline{D}_t)$ gives the holomorphic model of V: the action of H is given by

$$\begin{aligned} d \in D_t : &\quad f(x) \longmapsto f(x+d), \\ e \in \overline{D}_t : &\quad f(x) \longmapsto \exp(-2\pi i\langle e, x\rangle) f(x), \end{aligned}$$

and the Hilbert structure for which (f, f) is the integral of

$$f(d)\overline{f(d)}\exp(2\pi i\langle d, \overline{d}\rangle)$$

is invariant under $H_{\mathbb{R}}$; moreover $r_{\overline{D}_t} \circ \Omega_t$ identifies V with the space of f of finite norm. For $d = (-tz, z)$, the factor $\exp(2\pi i\langle d, \overline{d}\rangle)$, the commutator of d and $\overline{d}$, is

$$\exp(2\pi i\langle d, \overline{d}\rangle) = \exp(2\pi i(t - \overline{t})|z|^2).$$

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Fix E. The isomorphisms r_E of (2.1) make it possible to regard the W_D for $D \neq E$ as a family of representations of H operating in the fixed space $\mathcal{O}(E)$, with the action of H depending holomorphically on D. Let us make explicit the dependence on E of r_E. Choose coordinates as in no. 1, with E the x-axis. Let E' be generated by $(1, u)$ and D by $(-t, 1)$. We use x as coordinate on both E and E'. Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for $(1 + tu)x' = x$ and $\lambda = \exp(\pi iuxx')$. Hence $r_{E'}r_E^{-1}$ sends $f_E \in \mathcal{O}(E)$ to $f_{E'} \in \mathcal{O}(E')$, with $f_{E'}(x') = \lambda f_E(x)$:

$$f_{E'}\left(\frac{x}{1 + tu}\right) = \exp\left(\frac{\pi iu}{1 + tu}x^2\right) f_E(x). \tag{4.1}$$

This formula is holomorphic in t: the W_D form a holomorphic vector bundle $\mathcal{W}$ of infinite dimension over S, on which H acts.

To understand (4.1), one may observe that the point of E with coordinate x has the same image in $\mathbb{C}^2/D$ as the point of E' with coordinate $x/(1 + tu)$.

From (4.1) one deduces that the following growth condition on $v \in W_D$ is independent of the choices of $E \neq D$ and of the coordinate z on E:

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| \leq O\big(\exp(A|z|^2)\big). \tag{4.2}$$

Let $W_D^0 \subset W_D$ be defined by this condition. The W_D^0 form a holomorphic sub-bundle $\mathcal{W}^0$ of the bundle $\mathcal{W}$ in holomorphic representations of H.

5. Proposition

Let $T > 0$ and consider the following conditions on an entire function f:

(i)$_T$ There exists a holomorphic function $g(z, t)$, for $z \in \mathbb{C}$ and $|t| < T$, satisfying the heat equation $(\partial_t - \partial_z^2)g = 0$, with initial condition $g(z, 0) = f(z)$.

(ii)$_T$ $|f(z)| \leq O(\exp(|z|^2/4T))$.

(iii)$_T$ $\displaystyle\int f(z)\overline{f(z)}\exp(-|z|^2/2T)|\,dz \wedge d\overline{z}| < \infty$.

If $T < T_1$, each of $(i)_{T_1}$, $(ii)_{T_1}$, $(iii)_{T_1}$ implies each of $(i)_T$, $(ii)_T$, $(iii)_T$.

Proof. $(i)_{T_1} \Rightarrow (ii)_T$. The heat equation remains satisfied by the derivatives of g, and therefore

$$\partial_z^{2n} g = \partial_t^n g, \qquad \partial_z^{2n+1} g = \partial_t^n(\partial_z g).$$

Since $g(0, t)$ is holomorphic for $|t| < T_1$, if $T < T_2 < T_1$, then at $(0, 0)$

$$|\partial_t^n g/n!| \leq O(1/T_2^n), \qquad |\partial_z^{2n} g| \leq O(n!/T_2^n),$$

and there is an analogous estimate for $\partial_z^{2n+1} g$. The Taylor series of f then gives

$$|f(z)| \leq O\left(\sum \frac{n!}{(2n)!}\frac{1}{T_2^n}|z|^{2n}\right).$$

One has

$$\frac{1}{2n+1}2^{2n} \leq \frac{(2n)!}{(n!)^2} \leq 2^{2n},$$

hence

$$|f(z)| \leq O\left(\sum 2^{-2n}\frac{1}{T^n}\frac{|z|^{2n}}{n!}\right),$$

and this sum is $\exp(|z|^2/4T)$.

$(ii)_{T_1} \Rightarrow (i)_T$. It suffices to take g given by the kernel of the heat equation:

$$g(z,t) = \int_{\sqrt{t}\,\mathbb{R}} f(z+x)\frac{1}{2\sqrt{\pi t}}\exp(-x^2/4t)\,dx.$$

Changing $\sqrt{t}$ to $-\sqrt{t}$ changes the orientation of the cycle $\sqrt{t}\,\mathbb{R}$ and the sign of the integrand, so that the integral does not change. Condition (ii) ensures convergence for $|t| < T$.

The relation between (ii) and (iii) is clear: $(ii)_{T_1} \Rightarrow (iii)_T$, and $(iii)_{T_1} \Rightarrow (ii)_T$ is verified by writing $f(z)$ as the mean of its values on a disk with center z and radius $1/|z|$, and estimating that mean by Cauchy–Schwarz.

6

In homage to the Stone-von Neumann theorem, the bundle $\mathcal{W}$ over S is, projectively, equipped with a holomorphic connection for which the action of H is horizontal.

Let us specify "projectively." On the projective line S there is the ample invertible sheaf $\mathcal{O}(1)$. Let $\mathcal{O}(-1)$ be its dual. A square root $\mathcal{O}(-\frac{1}{2})$ of $\mathcal{O}(-1)$ is an invertible sheaf L endowed with an isomorphism $L^{\otimes 2} \simeq \mathcal{O}(-1)$. Locally such a square root exists and, locally, two square roots are isomorphic; but the isomorphism is not unique, with ambiguity ± 1. This ambiguity obstructs the global existence of $\mathcal{O}(-\frac{1}{2})$. It is not on $\mathcal{W}$, but on

$$\mathcal{W}(-\tfrac{1}{2}) := \mathcal{W} \otimes \mathcal{O}(-\tfrac{1}{2})$$

that there is a connection:

$$\nabla : \mathcal{W}(-\tfrac{1}{2}) \longrightarrow \mathcal{W}(-\tfrac{1}{2}) \otimes \Omega^1. \tag{6.1}$$

Note that if $\mathcal{O}(-\frac{1}{2})'$ and $\mathcal{O}(-\frac{1}{2})''$ are two square roots of $\mathcal{O}(-1)$, and if ∇ is a connection on $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})'$, then its transport to $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})''$ by an isomorphism $a : \mathcal{O}(-\frac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})''$ of square roots of $\mathcal{O}(-1)$ does not depend on the choice of a, which can be changed locally only by a constant ± 1. Thus the notion of a "connection on $\mathcal{W}(-\frac{1}{2})$" is well defined independently of the global existence, or the choice, of $\mathcal{O}(-\frac{1}{2})$.

Let $D \in S$ and let t be a tangent vector at D. For a local section s of $\mathcal{W}(-\frac{1}{2})$, the problem is to define $\nabla_t s$ in the fiber of $\mathcal{W}(-\frac{1}{2})$ at D. We proceed as follows: after making an auxiliary choice u, one defines $\nabla_t^u s$ for s a local section of $\mathcal{W}$ or of $\mathcal{O}(-\frac{1}{2})$. This derivation ∇_t^u depends on u, with dependence in u of the form

$$\nabla_t^{u'}(s) = \nabla_t^u(s) + as(D),$$

where a takes opposite values for $\mathcal{W}$ and for $\mathcal{O}(-\frac{1}{2})$. Hence, for $s = s_1 s_2$, with s_1 a local section of $\mathcal{W}$ and s_2 a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of u.

Let $D \in S$ be a line of $\mathbb{C}^2$. A tangent vector t at $D \in S$ is identified with a linear map $t : D \to \mathbb{C}^2/D$, i.e. with an element of $(\mathbb{C}^2/D)^{\otimes 2}$: to $u' \otimes v' \in (\mathbb{C}^2/D)^{\otimes 2}$ one associates $t(d) = \langle u, d\rangle v'$, where u is any representative of u'. The auxiliary datum will be a lift $\tilde{t} : D \to \mathbb{C}^2$ of t, or equivalently a lift of t to $\mathbb{C}^2/D \otimes \mathbb{C}^2$. We write it $\tilde{t} = u' \otimes u$, with $u \in \mathbb{C}^2$ mapping to u' in $\mathbb{C}^2/D$.

The tangent space at the origin of H is $\mathfrak{h} = \mathbb{C}^2 \times \mathbb{C}$. For z in this tangent space, let δ_z be the corresponding right-invariant vector field. One has

$$[\delta_x, \delta_y] = -\delta_{[x,y]},$$

where on the right appears the bracket in the Lie algebra of H. For $x, y \in \mathbb{C}^2$, computing $[x, y]$ as an infinitesimal commutator, one deduces

$$[\delta_x, \delta_y] = -2\pi i \langle x, y\rangle \delta_{(0,0,1)}. \tag{6.2}$$

Let $\mathcal{U}_1$ be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation $\delta_{(0,0,1)} = 1$. In $\mathcal{U}_1$, (6.2) implies that, for $d \in D$,

$$\left[-\frac{\delta_u^2}{4\pi i}, \delta_d\right] = \frac{1}{4\pi i} 2 \cdot 2\pi i \langle u, d\rangle \delta_u = \langle u, d\rangle \delta_u = \delta_{t(d)}. \tag{6.3}$$

Let us compute to first order around D. For $\varepsilon^2 = 0$ and $\varphi(g)$ in W_D, so $\delta_d\varphi = 0$, one has

$$\delta_{d+\varepsilon t(d)} \left[\varphi(g) + \varepsilon \frac{-\delta_u^2}{4\pi i}\varphi(g)\right] = \varepsilon \left[\delta_{t(d)}\varphi(g) + \delta_d \frac{-\delta_u^2}{4\pi i}\varphi(g)\right] = 0,$$

because $\delta_d(-\delta_u^2/4\pi i)\varphi = -[-\delta_u^2/4\pi i, \delta_d]\varphi$, by (6.3). To first order around D, the function of g and D' defined by

$$\varphi(g, D + \varepsilon t) = \varphi(g) + \varepsilon(-\delta_u^2/4\pi i)\varphi(g)$$

is therefore a section of the bundle $\mathcal{W}$. If $\varphi(g, D')$ is a local section of $\mathcal{W}$, this permits one to define the derivative $\nabla_t^u \varphi$ in $\mathcal{W}_D$ by

$$\nabla_t^u \varphi = \delta_t \varphi(g, D') + (\delta_u^2/4\pi i)\varphi(g, D). \tag{6.5}$$

Let us compute the dependence of (6.5) on u. If u is replaced by $u + x$ with $x \in D$, then in $\mathcal{U}_1$

$$\delta_{u+x}^2 = (\delta_u + \delta_x)^2 = \delta_u^2 + 2\delta_u\delta_x + [\delta_x, \delta_u] = \delta_u^2 + 2\delta_u\delta_x + 2\pi i \langle u, x\rangle. \tag{6.6}$$

For $\varphi \in W_D$, $\delta_x\varphi = 0$, and (6.6) becomes

$$\delta_{u+x}^2 \varphi = \delta_u^2 \varphi + 2\pi i \langle u, x\rangle \varphi, \tag{6.7}$$

whence

$$\frac{1}{4\pi i}\delta_{u+x}^2 \varphi = \frac{1}{4\pi i}\delta_u^2 \varphi + \frac{1}{2}\langle u, x\rangle \varphi. \tag{6.8}$$

The choice of u also makes it possible to define the derivative ∇_t^u with respect to t of a local section of $\mathcal{O}(1)$. A section of $\mathcal{O}(1)$ on $U \subset S$ is a homogeneous function of degree 1 on the inverse image of U in $\mathbb{C}^2$, and one differentiates with respect to $\langle u, d\rangle u$ to obtain a homogeneous function of degree 1 on D. When u is changed into $u + x$,

$$\nabla_t^{u+x}(s) = \langle u + x, d\rangle \delta_{u+x} s = \nabla_t^u(s) + \langle u, x\rangle s.$$

This derivation induces one on $\mathcal{O}(-\frac{1}{2})$, with

$$\nabla_t^u(s^{-1/2}) = -\frac{1}{2}\nabla_t^u(s)\, s^{-3/2},$$

and, for the dependence on u, s now being a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2}\langle u, x\rangle s. \tag{6.9}$$

The dependences on u in (6.8) and (6.9) cancel to provide a derivation ∇_t for $\mathcal{W}(-\frac{1}{2})$. Since the connection ∇ is defined in terms of right-invariant vector fields on H, it commutes with the action of H.

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Let us spell out the connection of no. 6 in coordinates. Choose coordinates as in no. 1, and let E be the x-axis. On S, $\mathcal{O}(1)$ has a section s with a simple zero at E. It is unique up to a factor and trivializes $\mathcal{O}(1)$ on $S - E$. Thus there is a square root $\mathcal{O}(-\frac{1}{2})$ on $S - E$, trivialized by a square root $s^{-1/2}$ of s^{-1}. By such a trivialization, the desired connection is transported to a connection on the bundle $\mathcal{W}$. We calculate it.

Let D_t be the line generated by $(-t, 1)$. By r_E of (2.1), a local section of $\mathcal{W}$ is identified with a function $f(z, t)$ holomorphic in z and t. We compute its derivative with respect to t. In no. 6, one may take

$$\tilde{t} : D_t \longrightarrow \mathbb{C}^2 : (-tz, z) \longmapsto (-1, 0)z; \quad \text{that is } d \longmapsto -\langle (1,0), d\rangle (1, 0).$$

The corresponding derivation of $\mathcal{O}(1)$ annihilates s, so that simply

$$\nabla_t f = (\partial_t - \partial_z^2/4\pi i)f, \tag{7.1}$$

and the horizontal local sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2/4\pi i)f = 0. \tag{7.2}$$

From no. 5 one deduces that through $s \in W_D^0(-\frac{1}{2})$ there passes a horizontal local section of $\mathcal{W}(-\frac{1}{2})$. It is automatically a section of $\mathcal{W}^0(-\frac{1}{2})$. It exists in a neighborhood of D that is larger the slower the growth of s. Note that in infinite dimension a connection does not necessarily define a local trivialization, even if through each point there passes a local horizontal section.

The connection ∇ on $\mathcal{W}^0(-\frac{1}{2})$ defines a connection on the corresponding bundle of projective spaces

$$\mathbb{P}(\mathcal{W}^0) = \mathbb{P}(\mathcal{W}^0(-\tfrac{1}{2})).$$

The horizontal local sections are the images of the nonzero horizontal sections of $\mathcal{W}^0(-\frac{1}{2})$. Although the sphere S is simply connected, one has the following.

8. Surprise

The projective-space bundle $\mathbb{P}(\mathcal{W}^0)$ on S has no global horizontal section.

The argument of no. 1 would have made it possible to predict this: apply it to the projective representation of $\mathrm{SL}(2, \mathbb{C})$ given by the action of $\mathrm{SL}(2, \mathbb{C})$ on the projective space of global horizontal sections of $\mathbb{P}(\mathcal{W}^0)$.

First proof. Divide S into two hemispheres S_1 and S_2, and choose $\mathcal{O}(-\frac{1}{2})_i$ on S_i. On $S_1 \cap S_2$ one has locally an isomorphism $\alpha : \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$, defined up to sign, but α can be defined globally only on the double cover of the circle $S_1 \cap S_2$: the monodromy is -1. If there existed global horizontal projective sections of $\mathbb{P}(\mathcal{W}^0)$, there would exist horizontal sections f_1, f_2 of $\mathcal{W}^0(-\frac{1}{2})$ on S_1 and S_2 which, on $S_1 \cap S_2$, would coincide up to sign. On $S_1 \cap S_2$

this would make it possible to normalize $\alpha : \mathcal{O}(-\frac{1}{2})_1 \to \mathcal{O}(-\frac{1}{2})_2$ by $\alpha(f_1) = f_2$, contradicting the nonexistence of α on $S_1 \cap S_2$.

Second proof. Choose coordinates as in no. 1. Let E be the x-axis and D the y-axis. As in no. 2, identify W_D with the space of entire functions $f(z)$. Let $f \neq 0$ in W_D. By the proposition, for there to exist on $S - E$ a horizontal section of $\mathbb{P}(\mathcal{W}^0)$ passing through f, it is necessary and sufficient that f satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| \leq O(\exp(\varepsilon |z|^2)). \tag{8.1}$$

If E' is another line, then by (4.1) the existence of a horizontal section on $S - E'$ passing through f is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| \leq O(\exp(\varepsilon |z|^2)) \tag{8.2}$$

with suitable $a \neq 0$.

Put $g(z) = f(z) \exp(az^2/2)$. By (8.1) and (8.2), $|g(z)|$ is bounded by a multiple both of $|\exp(\varepsilon|z|^2 + az^2/2)|$ and of $|\exp(\varepsilon|z|^2 - az^2/2)|$; hence in all directions except those for which az^2 is purely imaginary, g decreases exponentially quadratically. In all directions, it has uniformly at most exponential-quadratic growth. By Phragmén–Lindelöf, g decreases exponentially quadratically in all directions, hence is zero, and therefore $f = 0$.

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In the coordinates of no. 1, let D_t be the line generated by $(-t, 1)$. Let $d = (-tz, z)$. The commutator of d and $\overline{d}$ is

$$\langle d, \overline{d} \rangle = \exp(-2\pi i (t - \overline{t})|z|^2) \in \mathbb{C}^*;$$

the commutator is real positive, and

$$\langle d, \overline{d} \rangle > 1 \iff \operatorname{Im}(t) > 0.$$

The lines D such that, for d generating D, one has $\langle d, \overline{d} \rangle > 1$, therefore form a disk in S, called the disk attached to the real structure $H_{\mathbb{R}}$.

The real structures $L \subset \mathbb{C}^2$ for which $\langle \ , \ \rangle$ is real on L form a homogeneous space under $\mathrm{SL}(2, \mathbb{C})$, with $\mathrm{SL}(2, \mathbb{R})$ as the stabilizer of $\mathbb{R}^2$. Each defines a real structure $H_L = L \cdot U^1$ on H, hence a complex conjugation σ_L with fixed points LU^1.

The disks in S likewise form a homogeneous space under $\mathrm{SL}(2, \mathbb{C})$, the stabilizer of the one associated to $H_{\mathbb{R}}$ being $\mathrm{SL}(2, \mathbb{R})$. The correspondence

$$L \longmapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in S.

Passing from a disk to the complementary disk corresponds, on real structures, to $L \mapsto iL$.

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The calculations of no. 3 admit the following interpretation. Let X be an open disk of S, corresponding to a real structure L, whence real forms H_L of H and $\mathrm{SL}(L)$ of $\mathrm{SL}(2, \mathbb{C})$. Then $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla = 0})$ has an H_L-invariant pre-Hilbert structure; its completion $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge$ is the irreducible unitary representation of H_L. One has

$$\Gamma(X, \mathcal{W}^0(-\tfrac{1}{2})^{\nabla=0}) \subset \Gamma(X, \mathcal{W}^0(-\tfrac{1}{2})^{\nabla=0})^\wedge \subset \Gamma(X, \mathcal{W}(-\tfrac{1}{2})^{\nabla=0}).$$

The double cover $\mathrm{SL}(L)^\sim$ of $\mathrm{SL}(L)$ acts on $\mathcal{O}(-\frac{1}{2})|_X$ compatibly with its action on $\mathcal{O}(-1)$; hence $\mathrm{SL}(L)^\sim$ acts on $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$. This is the action that normalizes the action of H_L and induces the natural action of $\mathrm{SL}(L)$ on H_L.

The subrepresentation $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ may be interpreted as the space of real analytic vectors for the action of $\mathrm{SL}(L)^\sim$.

This description shows that the projective representation of $\mathrm{SL}(L)$ extends to the monoid of $g \in \mathrm{SL}(2, \mathbb{C})$ such that $gX \supset X$.

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Everything preceding generalizes from $\mathrm{SL}(2, \mathbb{C})$ to $\mathrm{Sp}(2n, \mathbb{C})$: replace $\mathbb{C}^2$ by $\mathbb{C}^{2n}$, endowed with a nondegenerate alternating form $\langle\ ,\ \rangle$ real on $\mathbb{R}^{2n}$, replace H by $\mathbb{C}^{2n} \times \mathbb{C}^*$ with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y\rangle)),$$

and replace S by the space of Lagrangian, i.e. maximal isotropic, subspaces D of $\mathbb{C}^{2n}$. The disk attached to a real structure becomes the Siegel space of those D for which $|\langle d, \overline{d}\rangle| > 1$ for $d \neq 0$ in D. The invertible sheaf $\mathcal{O}(-\frac{1}{2})$ must be replaced by $\det(D)^{1/2}$.

12. Remark

Beside representations of H in which $\mathbb{C}^*$ acts by multiplication by λ, consider likewise representations in which the action is by λ^{-1}. One defines, as in no. 2, a holomorphic bundle $\mathcal{W}^0_-$ of such representations over S, and on $\mathcal{W}^0_-(-\frac{1}{2})$ one again has a connection, given by a heat equation.

On a connected simply connected open subset U of S, consider $W = \mathcal{W}^0(-\frac{1}{2})$, its connection, and the action of H. In $\Gamma(U, W)^{\nabla=0}$, for each $x \in U$ there is a linear form fixed by the corresponding line, defined up to a factor: $f \mapsto f(e)$ for $f \in \mathcal{W}^0_x$. For each $x \notin U$, there exists a vector $v(x)$ fixed by D_x; in the Schrödinger model, and for x corresponding to $E =$ the x-axis, it is the constant 1. It is unique up to a factor.

Fix the real structure $H_\mathbb{R}$, whence a real structure $S_\mathbb{R} \subset S$ (a circle), and a decomposition of $S - S_\mathbb{R}$ into two disks U_1, U_2. Then:

(a) The complex conjugate of $\Gamma(U_1, W)^{\nabla=0}$ is $\Gamma(U_2, W_-)^{\nabla=0}$.

(b) The Hermitian form on $\Gamma(U_1, W)^{\nabla=0}$, invariant under $H_\mathbb{R}$, is reinterpreted as a bilinear pairing between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, invariant under H.

Question. Let C be a Jordan curve in S, separating $S - C$ into two open sets U_1 and U_2. Does there exist a unique separating H-invariant bilinear form pairing $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, induced by a perfect duality between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$?

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